

Jenny Liu

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Intended major in Mechanical Engineering

Aug 2026 - Expected May 2030

TECHNICAL EXPERIENCE

Valiant TMS, Windsor, Canada, *Industrial Design Group Intern*

Jun 2024 - Jul 2024

- Created preliminary 3D models from engineering drawings in Autodesk Inventor for existing automation projects
- Participated in and documented 3 supplier design meetings involving conveyor belts, shaft collars, and flywheels
- Collaborated with the programming team to test FANUC robot motion using ROBOGUIDE and TP code

Robotics Club, Branksome Hall High School, *Club Head*

Sep 2024 - May 2026

- Organized and led meetings with 20+ robotics club members for 6 hours weekly
- Designed and built VEX mechanisms through CAD, prototyping, and competition testing
- Participated in 5 VEX competitions annually and earned the VEX Judges Award and Semifinalist recognition in the RECF Community Challenge, presented by Google

World Affairs Conference, *Technology Department Head*

Sep 2024 - Mar 2026

- Maintained and enhanced a 1,500+ line JavaScript, HTML, and CSS web application for WAC, a student-led conference engaging high-school students with global issues
- Integrated Firebase for student-teacher registration and managed issue tracking through GitHub Projects
- Supported technology operations for a conference serving 1,200+ participants from 15+ Toronto schools

RESEARCH PROJECTS

“SNAP-HAR: Signal-Neural Adaptive Processing for Robust Human Activity Recognition” May 2025 - Jul 2025

Independent Researcher - CONF-MLA 2025

- Developed a machine-learning framework for robust wearable-sensor activity recognition in Python
- Designed experiments across 3 datasets and 5 masking strategies to evaluate model performance
- Improved F1-score by 17.0% on clean data and 24.6% on noisy data

“Analysis of the Driving Stability of a Multi-Model Drive Electric-Tracked Intelligent Detection Device”

Jilin University, *High School Research Intern* - Actuators, 13(12), 2024

Apr 2024 - Nov 2024

- Co-authored a peer-reviewed paper in Actuators following a 3-month research internship under Prof. Zhenglei Yu
- Developed a RecurDyn virtual prototype of an electric-tracked detection device and used MATLAB to process and analyze simulation results
- Conducted physical prototype testing across 3 operating conditions and analyzed forward, vertical, and roll angular acceleration to evaluate driving stability

Emergency Door Release Mechanism in Electric Vehicle Fires, *Independent Researcher*

Nov 2023 - Mar 2024

- Designed and prototyped an emergency EV door-release mechanism in SolidWorks integrating temperature sensing, solenoid actuation, and a pneumatic release system
- Conducted 65 trials across 13 test conditions to evaluate reliability and analyzed experimental data using Python
- Earned a Gold Medal at the Toronto Science Fair and presented the project at a school-wide STEM assembly

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, HTML/CSS, FANUC TP

Simulation & Analysis: RecurDyn, MATLAB, FANUC ROBOGUIDE

Mechanical Design: SolidWorks, Autodesk Inventor, 3ds Max, Blender

Languages: English (bilingual); Mandarin (bilingual); Spanish (introductory)